

Program : Diploma in Manufacturing Technology	
Course Code : 4701	Course Title: Mechanical Engineering
Semester : 4	Credits: 4
Course Category: Program Core	
Periods per week: 4 (L:3, T:1, P:0)	Periods per semester: 60

Course Objectives:

- To provide students to get a good knowledge about the importance of studying Mechanical engineering in daily life and industry.
- To familiarize with the basics of thermal engineering.
- To familiarize with the basics of Refrigeration and Air conditioning

Course Prerequisites:

Topic	Course Code	Course Name	Semester
Basic Science		Applied physics I Applied physics II	1 & 2
Basic Science		Mathematics I Mathematics II	1 & 2

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Explain the concepts in thermodynamics, thermodynamic laws and air standard cycles	20	Applying
CO2	Explain the properties of steam, working of steam turbines, petrol and diesel engines	14	Applying
CO3	Describe different modes of heat transfer, types of heat exchangers and working of air compressors	15	Applying

CO4	Describe the principles of refrigeration, psychrometry and air conditioning process.	9	Applying
	Series Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3	2		3			
CO3	3						
CO4	3	2		3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Express the basics of thermodynamics, different thermodynamic processes and air standard cycles		
M1.01	Identify and define the terms in thermodynamics	3	Understanding
M1.02	State different gas laws	3	Applying
M1.03	State and explain thermodynamics laws	4	Applying
M1.04	Explain different thermodynamic processes and illustrate with pV and Ts diagram	5	Applying
M1.05	Explain different Air standard cycles and Illustrate with P-V, T-S diagram.	5	Applying
Contents: Define and classify system- boundary - surrounding- heat and work.- intrinsic and extrinsic properties- Thermodynamic equilibrium-Quasistatic processes-Define- ,temperature, enthalpy, entropy and their SI units.- State Boyle's law, Charles's law, Joule's law.-Specific heats-ratio of specific heats-State characteristic gas equation., mayors relation.-simple problems. Thermodynamic processes- Isochoric, Isobaric, Isothermal, adiabatic, Polytrophic and throttling processes with pV and Ts diagrams. Simple problems using the relation for			

calculating work done, Internal energy. Thermodynamic laws- Zeroth law, first law, and second law of thermodynamics(statements only) Air standard cycles- assumptions- air standard efficiency. -Illustrate with pV and Ts diagram. Carnot cycle, Otto cycle, Diesel cycle.-simple problems on Otto and Carnot cycle.			
CO2	Identify the Properties of steam,working of steam turbines, petrol engines and diesel engines		
M2.01	Explain the formation of steam at constant pressure and illustrate it with a graph	4	Understanding
M2.02	Identify and Apply steam table to compute the enthalpy of wet, dry and superheated steam	5	Applying
M2.03	Appreciate the working of steam, petrol and diesel engine. Estimate IP, BP and efficiency of engines.	5	Applying
	Series Test – I	1	
Contents: Steam and its properties: Generation of steam- Distinguish between wet, dry, superheated steam-Compute the enthalpy of wet, dry and super heated steam at the given pressure and state using steam table- Determine conditions of steam- enthalpy, entropy, specific volume of steam using mollier chart-Simple problems-Working of steam turbines with line diagram-Working of petrol and diesel engine (two stroke and four stroke).-Brake power, indicated power and friction power efficiency of engines. Comparison between SI and CI engines.			
CO3	Understanding different modes of heat transfer, types of heat exchangers and compressors		
M3.01	Identify different modes of heat transfer and state Fourier's law	3	Understanding
M3.02	Discuss Conduction through a plane wall and solve simple problems.	3	Applying
M3.03	Identify different types of heat exchangers and their effectiveness	3	Understanding
M3.04	Explain the working of single stage air compressor.	3	Understanding
M3.05	Discuss the classification and working of different types of air compressors.	3	Understanding
Contents: Heat transfer- modes of heat transfer-conduction-convection and radiation. Fourier's law of thermal conduction.-Thermal conductivity-conduction through a plane wall -simple problems.			

Heat exchangers -Classification- parallel flow- counter flow type – shell and tube – effectiveness of a heat exchanger(definition only) Air compressors -classification- Uses of compressed air.-Single stage reciprocating compressor its construction and working (with line diagram) using pV diagram. Rotary compressors: centrifugal compressors, Axial flow type compressors, vane type compressors, lobe type compressors (working principle with line diagram.)			
CO4	Describe the principles of refrigeration, psychrometry and air conditioning process.		
M 4.01	Ability to define terms related to refrigeration	1	Applying
M 4.02	Understand principle and types of refrigeration	1	Understanding
M 4.03	Identify basic components of air and vapour compression refrigeration system	1	Understanding
M 4.04	Identify different terms related to Psychrometry	2	Understanding
M 4.05	Ability to solve simple problems using psychrometric chart	2	Applying
M 4.06	Ability to define air conditioning and identify the factors affecting human comfort	2	Understanding
	Series Test – II	1	
Contents: Introduction to Refrigeration: Definition of Refrigeration; Refrigerating effect-unit of refrigeration- Coefficient of performance-refrigerant. Types of Refrigeration -Carnot refrigeration Cycle, Principle of air refrigeration-Advantages and disadvantages in air refrigeration; - Principle of Vapour compression refrigeration system. Psychrometry -Definition- Dry air- moist air- saturated- unsaturated -super saturated air-degree of saturation- dry bulb temperature- wet bulb temperature- dew point temperature-Dalton's law of partial pressures -absolute humidity-relative humidity- specific humidity-Enthalpy of moist air Psychrometricprocess-sensible heating, sensible cooling, humidification and dehumidification-representation of psychrometric processes on psychrometric chart. Air conditioning: Air Conditioning-Definition-Terms- factors affecting human comfort-working principle of room air conditioner.			

Text / Reference

T/R	Book Title/Author
T1	A text book of Thermal Engineering -R.S.Khurmi&J.K.Gupta
T2	Thermal Engineering -D.S.Kumar

T3	Refrigeration and Air conditioning- Arora
R4	Thermal Engineering -P.L.Ballany
R5	Thermodynamics- P K Nag
R6	Internal combustion engine- M L Mathur and R P sharma

Online Resources

Sl.No	Website Link
1	https://lecturenotes.in/video-tutorial/15813-note-of-thermal-engineering-1-video-tutorial-by-bhaktishree-sathpathy
2	https://onlinecourses.nptel.ac.in/noc20_me71/preview